

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer **ALL** questions in this section. Marks are indicated at the end of each question. Together they are worth 80% of the total marks for this examination.

Question 1 (21 marks – approximately 38 minutes)

- (a) List and describe any **THREE** quality-related characteristics of data and information.
(6 marks)
- (b) State and describe the **THREE** roles of information systems.
(6 marks)
- (c) Outline any **THREE** types of e-commerce activities that could be done by an online fashion store.
(3 marks)
- (d) More and more businesses set up websites and mobile applications to provide self-service functions to their customers.

Demonstrate and describe how the following industries provide conveniences to their customers with self-service functions:

- (i) airline;
- (ii) catering; and
- (iii) banking.

(6 marks)

Question 2 (18 marks – approximately 32 minutes)

- (a) Many companies and organisations apply different information technologies to their businesses and operations to improve their productivity and operating efficiency. Hence, accountants nowadays are not solely data users and data creators anymore.

Explain how accountants perform the following roles:

- (i) system designers; and
- (ii) system auditors.

(6 marks)

- (b) Demonstrate any TWO types of information that an executive needs in each of the following departments:

- (i) human resources; and
- (ii) production.

(6 marks)

- (c) An efficient information system can provide relevant information to an organisation's management, allowing the management to know how to develop new strategies and improve performance.

State any TWO examples of how an information system could enhance the activities below:

- (i) continuous process improvement; and
- (ii) organisational communication.

(6 marks)

Question 3 (14 marks – approximately 25 minutes)

Williams Toys Manufacturing Company ("Williams") has used its canned accounting software to handle accounting, sales and purchasing transactions for over ten years. Six years ago, Williams started to use spreadsheets to handle production plans and material consumption. The management of Williams is now exploring whether to acquire a new integrated enterprise resource planning ("ERP") system to replace its existing canned accounting software and spreadsheet operations.

Required:

- (a) State and describe any TWO reports detailing how the management of Williams can monitor the performance of its canned accounting software. (4 marks)

One day, a preliminary quotation of the ERP system is received with the following proposed professional services:

Total number of consultation days: 20

Particulars:

Existing system review and meeting with users	4 days
Layout customisation among reports and documents	8 days
System training	8 days

Required:

- (b) Identify a stage of the systems development life cycle for each professional service mentioned above. (3 marks)
- (c) Explain why a user acceptance test should be completed before an ERP system is officially launched. (4 marks)
- (d) Demonstrate any THREE activities that the management of Williams may do in selecting a suitable ERP system. (3 marks)

Question 4 (11 marks – approximately 20 minutes)

- (a) Define artificial intelligence ("AI").
(2 marks)
- (b) Contrast the difference between Strong AI and Weak AI.
(3 marks)
- (c) State any THREE capabilities of a performance dashboard.
(3 marks)
- (d) A dashboard is commonly used to monitor key performance metrics of a company or an organisation.

Demonstrate and describe how a business intelligence system presents the following matters with a dashboard:

- (i) an increase of net profit ratio; and
- (ii) shortage of an inventory item.

(3 marks)

Question 5 (16 marks – approximately 29 minutes)

- (a) Outline and describe any THREE components of security policies. (6 marks)
- (b) Classify and describe any THREE common classes of data among non-governmental organisations. (6 marks)
- (c) Demonstrate and describe any TWO limitations of using passwords for authentication. (4 marks)

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